

Heavy-Tail Diffusion Factor Model

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1 Background

A factor model is a way to describe many asset returns through a small number of common sources of variation. Following Diffusion Factor Models (DFM) [Chen et al., 2026], let $\mathbf{R} \in \mathbb{R}^d$ denote the return vector of d assets. The factor model assumes

$$\mathbf{R} = \beta \mathbf{F} + \varepsilon,$$

where $\mathbf{F} \in \mathbb{R}^k$ is the vector of k unobserved common factors with $k \ll d$, $\beta \in \mathbb{R}^{d \times k}$ is the loading matrix that maps factor movements into asset returns, and $\varepsilon \in \mathbb{R}^d$ is the vector of idiosyncratic residuals. Here, a common factor is a latent variable that affects many assets at the same time, while an idiosyncratic residual is the asset-specific part of the return that is not explained by the common factors. Because $\mathbf{F}$ and β are not separately identifiable without a normalization, DFM normalizes β to have orthonormal columns, meaning that the columns of β have unit length and are mutually orthogonal. Its main score-decomposition result assumes that $\mathbf{F}$ and ε are independent and that ε is Gaussian with asset-specific variances.

DFM inserts this factor structure into a score-based diffusion model. A forward Ornstein-Uhlenbeck (OU) process gradually adds Gaussian noise to the return vector $\mathbf{R}$, producing a noisy return $\mathbf{R}_t$ at time t with density p_t . The score at time t is $\nabla \log p_t(\mathbf{r})$, the gradient of the log-density evaluated at a noisy return value $\mathbf{r}$; it tells the reverse diffusion process which direction increases data likelihood. DFM learns this score and then runs the reverse process to transform Gaussian noise into synthetic high-dimensional returns. Its key structural observation is that, under the factor model, the score decomposes into a subspace score and a complement score: the subspace score depends on a k -dimensional factor coordinate, while the complement score is linear and adjusts the covariance generated by the idiosyncratic residuals. This decomposition motivates a neural score network, meaning a neural network used to approximate $\nabla \log p_t$. The proposed architecture uses denoising to learn how to remove the added Gaussian noise, an encoder-decoder structure to map noisy returns into and out of a lower-dimensional representation, and skip connections to pass linear residual information around nonlinear layers. The resulting statistical guarantees are driven mainly by the intrinsic factor dimension k , rather than by the ambient asset dimension d , which is the main reason the method is useful in high-dimensional and small-sample financial settings.

This note starts from DFM and considers two possible extensions.

- 1. Factor models with heavy-tailed residuals.** The first direction is a factor model with heavy-tailed residuals, where “heavy-tailed” means that the residual distribution places more probability on extreme asset-specific shocks than a Gaussian distribution does. In that setting, the main questions are learning theory, namely finite-sample guarantees for estimating the score, and sampling theory, namely guarantees that the generated return distribution is close to the true heavy-tailed distribution. Recent work on heavy-tailed diffusion and flow methods is directly relevant: Pandey et al. [2024] introduce Student- t -based diffusion and flow variants, where a Student- t distribution is a standard heavy-tailed alternative to a Gaussian distribution; Yu and Yu [2026] study score estimation and sampling guarantees for heavy-tailed targets; and Guan et al. [2025] use heavy-tailed priors in flow matching, a generative method that learns a velocity field transporting a simple prior distribution, meaning the distribution from which generation starts, to a target distribution.
- 2. Learning factors through score networks.** The second direction keeps the original score-network viewpoint but asks whether the network can learn the loading structure itself. In the simplified formulation below, the period-specific loading matrix is written as $\beta_i = \mathbf{U}_i \mathbf{T}$, where $\mathbf{U}_i$ is a known descriptor

matrix and $\mathbf{T}$ is a time-invariant weight matrix. We then build a score network with a trainable matrix $\mathbf{A}$ in the same position as $\mathbf{T}$, so that learning the score may also learn the descriptor-to-factor loading map. This question is related to theory showing that score networks can recover unknown low-dimensional linear subspaces [Chen et al., 2023] and that diffusion training can be interpreted as subspace clustering under mixture-of-low-rank-Gaussian structure [Wang et al., 2024]. The main questions are identifiability and estimation: under what normalization can $\mathbf{T}$ or the loading space of $\mathbf{U}_i\mathbf{T}$ be recovered, and how accurately can the learned factors support downstream portfolio or risk tasks?

2 Preliminary

Notation. All vectors and matrices are written in boldface. For example, $\mathbf{R}$ denotes a return vector, $\mathbf{F}$ denotes a factor vector, and $\boldsymbol{\beta}$ denotes a loading matrix. Scalars, such as the number of assets d , the number of factors k , and time t , are not bold. For a vector $\mathbf{v}$, $\|\mathbf{v}\|_2$ and $\|\mathbf{v}\|_\infty$ denote its Euclidean norm and max norm, respectively. For a matrix $\mathbf{M}$, $\text{tr}(\mathbf{M})$, $\|\mathbf{M}\|_F$, and $\|\mathbf{M}\|_{\text{op}}$ denote its trace, Frobenius norm, and operator norm, respectively. If $\mathbf{M}$ is symmetric, $\lambda_{\max}(\mathbf{M})$ denotes its largest eigenvalue and $\lambda_j(\mathbf{M})$ denotes its j -th largest eigenvalue. For a positive semidefinite matrix $\mathbf{M}$, the matrix-weighted norm is $\|\mathbf{v}\|_{\mathbf{M}}^2 = \mathbf{v}^\top \mathbf{M} \mathbf{v}$. For two symmetric matrices $\mathbf{M}$ and $\mathbf{N}$, the notation $\mathbf{M} \succeq \mathbf{N}$ means that $\mathbf{M} - \mathbf{N}$ is positive semidefinite. For a random vector $\mathbf{X}$ with distribution P , we write $\|\mathbf{X}\|_{L^2(P)}^2 = \mathbb{E}[\|\mathbf{X}\|_2^2]$. We denote by $\phi(\cdot; \boldsymbol{\mu}, \boldsymbol{\Sigma})$ the Gaussian density with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. Throughout the note, uppercase bold letters, such as $\mathbf{R}$, denote random vectors, while lowercase bold letters, such as $\mathbf{r}$, denote their realizations.

2.1 Factor Models

Let $\mathbf{R} \sim P_{\text{data}}$ be the return vector of d assets, where P_{data} denotes the data-generating distribution of asset returns. In applications, we observe a finite sample $\{\mathbf{r}_i\}_{i=1}^n$, where each $\mathbf{r}_i \in \mathbb{R}^d$ is one realized return vector and n is the number of observations. Factor models assume that most of the common movement in $\mathbf{R}$ can be explained by a much lower-dimensional vector of latent variables. Following DFM, we write

$$\mathbf{R} = \boldsymbol{\beta}\mathbf{F} + \boldsymbol{\varepsilon}, \quad (1)$$

where $\mathbf{F} \in \mathbb{R}^k$ is the vector of k latent factors with $k \ll d$, $\boldsymbol{\beta} \in \mathbb{R}^{d \times k}$ is the loading matrix, and $\boldsymbol{\varepsilon} \in \mathbb{R}^d$ is the idiosyncratic residual vector. The j -th component of $\mathbf{F}$ is a common source of variation shared by many assets, while the i -th row of $\boldsymbol{\beta}$ records how the i -th asset responds to those k factors. The residual vector $\boldsymbol{\varepsilon}$ collects the asset-specific part of returns that is not explained by the common factors. This low-dimensional structure is a standard way to model systematic variation in asset returns, going back to arbitrage pricing ideas [Ross, 2013] and appearing in modern high-dimensional factor estimation and asset-pricing work [Fan et al., 2016, Ait-Sahalia and Xiu, 2019, Giglio and Xiu, 2021, Kelly et al., 2023].

The factors are unknown in DFM: the training data consist only of returns $\{\mathbf{r}_i\}_{i=1}^n$, not paired observations of $\mathbf{F}$ or $\boldsymbol{\varepsilon}$. This is important because many useful financial factors are latent, omitted, or estimated from noisy characteristics rather than directly observed [Giglio and Xiu, 2021, Bryzgalova et al., 2023]. The object that is identifiable from the return distribution is therefore not a unique pair $(\boldsymbol{\beta}, \mathbf{F})$, but the low-dimensional factor space. We define the factor space as

$$\mathcal{S}_{\boldsymbol{\beta}} := \{\boldsymbol{\beta}\mathbf{f} : \mathbf{f} \in \mathbb{R}^k\} \subseteq \mathbb{R}^d,$$

the set of all return vectors that can be generated by changing the factor realization $\mathbf{f}$ while setting the residual to zero. Equivalently, $\mathcal{S}_{\boldsymbol{\beta}}$ is the column space of $\boldsymbol{\beta}$. If $\mathbf{A} \in \mathbb{R}^{k \times k}$ is any invertible matrix, then $\boldsymbol{\beta}\mathbf{F} = (\boldsymbol{\beta}\mathbf{A})(\mathbf{A}^{-1}\mathbf{F})$. Thus the same factor component can be represented by different loading matrices and factor coordinates. This rotation and rescaling ambiguity is the usual identifiability issue in latent factor models.

To remove this ambiguity without changing the model class, we follow the normalization used in DFM and assume that the loading matrix has orthonormal columns:

$$\boldsymbol{\beta}^\top \boldsymbol{\beta} = \mathbf{I}_k,$$

where $\mathbf{I}_k \in \mathbb{R}^{k \times k}$ is the identity matrix. This normalization means that the k columns of $\boldsymbol{\beta}$ have unit length and are mutually orthogonal. It is without loss of generality when $\boldsymbol{\beta}$ has full column rank: one may write $\boldsymbol{\beta} = \tilde{\boldsymbol{\beta}}\mathbf{H}$, where $\tilde{\boldsymbol{\beta}} \in \mathbb{R}^{d \times k}$ has orthonormal columns and $\mathbf{H} \in \mathbb{R}^{k \times k}$ is invertible, and then absorb $\mathbf{H}$ into the factor vector by replacing $\mathbf{F}$ with $\mathbf{H}\mathbf{F}$. Under this convention, the model separates the high-dimensional ambient space $\mathbb{R}^d$ from the intrinsic factor dimension k , which is the dimension that DFM aims to exploit.

2.2 Diffusion Processes

We next recall the diffusion process used to turn the factor model into a generative model. Let $(\mathbf{W}_t)_{t \geq 0}$ be a standard Brownian motion in $\mathbb{R}^d$, meaning a continuous-time Gaussian noise process whose increments are independent and have covariance proportional to elapsed time. Starting from a return vector $\mathbf{R}_0 \sim P_{\text{data}}$, the standard Ornstein-Uhlenbeck forward process is

$$d\mathbf{R}_t = -\frac{1}{2}\mathbf{R}_t dt + d\mathbf{W}_t, \quad t \in [0, T],$$

where $T > 0$ is the terminal time. This process gradually shrinks the original return vector toward zero while adding Gaussian noise. Let P_t denote the distribution of $\mathbf{R}_t$, and let p_t denote its density whenever the density exists. Conditional on an initial realization $\mathbf{R}_0 = \mathbf{r}_0$, the forward process has the explicit Gaussian transition

$$\mathbf{R}_t \mid \mathbf{R}_0 = \mathbf{r}_0 \sim N(\alpha_t \mathbf{r}_0, h_t \mathbf{I}_d), \quad \alpha_t = e^{-t/2}, \quad h_t = 1 - e^{-t},$$

where $\mathbf{I}_d \in \mathbb{R}^{d \times d}$ is the identity matrix. For large T , P_T is close to the standard Gaussian distribution $N(\mathbf{0}, \mathbf{I}_d)$, so the forward process maps data into an easy-to-sample noise distribution.

Generation uses the time-reversed process. The score function at time t is

$$\mathbf{s}^*(\mathbf{r}, t) := \nabla \log p_t(\mathbf{r}),$$

where ∇ is the gradient with respect to the vector input $\mathbf{r}$. If the score function were known exactly and the reverse process were initialized from $\mathbf{R}_0^\leftarrow \sim P_T$, then the reverse-time SDE is

$$d\mathbf{R}_t^\leftarrow = \left\{ \frac{1}{2}\mathbf{R}_t^\leftarrow + \mathbf{s}^*(\mathbf{R}_t^\leftarrow, T - t) \right\} dt + d\mathbf{W}_t, \quad t \in [0, T].$$

The superscript $\leftarrow$ indicates that this process runs from noise back to data. Under the usual regularity conditions for diffusion time reversal, $\mathbf{R}_T^\leftarrow$ has distribution P_{data} . In practice, P_T and $\mathbf{s}^*$ are unknown, so one starts from $N(\mathbf{0}, \mathbf{I}_d)$ and trains a neural score network $\mathbf{s}_\theta(\mathbf{r}, t)$ to approximate $\mathbf{s}^*(\mathbf{r}, t)$. A common training objective is denoising score matching:

$$\hat{L}(\mathbf{s}_\theta) = \frac{1}{n} \sum_{i=1}^n \frac{1}{T - t_0} \int_{t_0}^T \mathbb{E} \left[\left\| \mathbf{s}_\theta(\mathbf{R}_t, t) + \frac{\mathbf{R}_t - \alpha_t \mathbf{r}_i}{h_t} \right\|_2^2 \mid \mathbf{R}_0 = \mathbf{r}_i \right] dt,$$

where $t_0 > 0$ is a small early-stopping time that avoids the singular behavior of the score near $t = 0$. The term $-(\mathbf{R}_t - \alpha_t \mathbf{r}_i)/h_t$ is the conditional Gaussian score of $\mathbf{R}_t$ given $\mathbf{R}_0 = \mathbf{r}_i$, so the loss trains $\mathbf{s}_\theta$ to denoise the noised return vector.

Heavy-Tailed Diffusion Processes. The above construction is Gaussian because the driving noise is Brownian motion and the perturbation kernel is Gaussian. A heavy-tailed diffusion process aims to better represent rare or extreme events by replacing some part of this Gaussian construction with a heavy-tailed object, such as a Student- t perturbation or prior [Pandey et al., 2024, Guan et al., 2025], an α -stable Lévy process with jumps [Yoon et al., 2023], or a conventional Gaussian diffusion analyzed under a heavy-tailed target distribution [Yu and Yu, 2026]. Here, an α -stable Lévy process is a stochastic process whose increments can have much heavier tails than Gaussian increments and may include jumps rather than continuous Brownian paths. This direction is promising for financial returns, where extreme residual shocks are common, but it is less developed than the standard Gaussian diffusion theory. In particular, the training objectives are not yet as canonical as the Gaussian diffusion loss: they may involve tailored denoising rules, fractional score matching, or robust divergences rather than the usual likelihood or evidence lower bound interpretation associated with classical diffusion models. As a result, both the learning theory and the sampling theory for heavy-tailed diffusion remain more problem-specific and less unified.

2.3 DFM

DFM combines the factor model (1) with the OU diffusion process described above. The key reason this combination is natural is that the factor model identifies a low-dimensional structure inside the high-dimensional return vector. When (1) holds under the loading normalization above, most nonlinear dependence in the data distribution is carried by the k -dimensional factor vector $\mathbf{F}$, while the idiosyncratic residual vector $\boldsymbol{\varepsilon}$ accounts for asset-specific noise in $\mathbb{R}^d$. A diffusion model must learn the score $\mathbf{s}^*(\mathbf{r}, t) = \nabla \log p_t(\mathbf{r})$ in the full ambient dimension d . Without structure, this is a high-dimensional nonparametric learning problem. The factor model gives additional geometry: the important nonlinear part of the score should live near the k -dimensional factor space $\mathcal{S}_\beta$, with $k \ll d$.

In the baseline setting of DFM, the factor vector $\mathbf{F}$ has a density p_{fac} and a finite second moment, and the residual vector is Gaussian:

$$\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \mathbf{D}), \quad \mathbf{D} := \text{diag}\{\sigma_1^2, \dots, \sigma_d^2\}.$$

where $\sigma_i^2 > 0$ is the idiosyncratic variance of the i -th asset, and $\mathbf{F}$ is independent of $\boldsymbol{\varepsilon}$. Under the OU forward process, the residual covariance and the added Gaussian noise combine into the time-dependent diagonal matrix

$$\boldsymbol{\Lambda}_t := h_t \mathbf{I}_d + \alpha_t^2 \mathbf{D} = \text{diag}\{h_t + \alpha_t^2 \sigma_1^2, \dots, h_t + \alpha_t^2 \sigma_d^2\}.$$

The matrix $\boldsymbol{\Lambda}_t \in \mathbb{R}^{d \times d}$ records the variance of the noisy residual component at time t . Define

$$\boldsymbol{\Gamma}_t := (\boldsymbol{\beta}^\top \boldsymbol{\Lambda}_t^{-1} \boldsymbol{\beta})^{-1}, \quad \mathbf{T}_t := \boldsymbol{\Lambda}_t^{-1/2} \boldsymbol{\beta} \boldsymbol{\Gamma}_t \boldsymbol{\beta}^\top \boldsymbol{\Lambda}_t^{-1/2}.$$

Here, $\boldsymbol{\Gamma}_t \in \mathbb{R}^{k \times k}$, and $\mathbf{T}_t \in \mathbb{R}^{d \times d}$ is the orthogonal projection onto the time-dependent rescaled factor subspace spanned by the columns of $\boldsymbol{\Lambda}_t^{-1/2} \boldsymbol{\beta}$. This time dependence matters because the OU noise level changes with t , and heterogeneous residual variances make different asset directions scale differently.

With these definitions, the score admits the decomposition

$$\begin{aligned} \mathbf{s}^*(\mathbf{r}, t) = & \underbrace{\mathbf{T}_t \boldsymbol{\Lambda}_t^{1/2} \boldsymbol{\beta} \nabla \log p_{\text{fac}, t} \left(\boldsymbol{\beta}^\top \boldsymbol{\Lambda}_t^{1/2} \mathbf{T}_t \boldsymbol{\Lambda}_t^{-1/2} \mathbf{r} \right)}_{\text{Subspace score}} \\ & - \underbrace{\boldsymbol{\Lambda}_t^{-1/2} (\mathbf{I}_d - \mathbf{T}_t) \boldsymbol{\Lambda}_t^{-1/2} \mathbf{r}}_{\text{Complement score}}. \end{aligned} \tag{2}$$

The density $p_{\text{fac}, t}$ is the Gaussian-smoothed factor density

$$p_{\text{fac}, t}(\mathbf{z}) := \int_{\mathbb{R}^k} \phi(\mathbf{z}; \alpha_t \mathbf{f}, \boldsymbol{\Gamma}_t) p_{\text{fac}}(\mathbf{f}) d\mathbf{f}, \quad \mathbf{z} \in \mathbb{R}^k.$$

Thus the first term in (2) is a subspace score: it requires estimating a nonlinear function of a k -dimensional factor coordinate through the Gaussian-smoothed factor density and then maps that score back to $\mathbb{R}^d$. The second term in (2) is a complement score: it is linear in the return realization $\mathbf{r}$ and adjusts the residual directions outside the factor subspace. This is the central score decomposition result in DFM.

The decomposition in (2) explains why the specified factor model is well suited for diffusion modeling. Instead of asking a neural network to learn an arbitrary nonlinear score on $\mathbb{R}^d$, the model suggests learning a low-dimensional nonlinear score together with a simple high-dimensional linear correction. This matches the financial intuition that a few common factors drive systematic return movements, while residual directions mostly contribute asset-specific variation. It also explains the architecture used in DFM: an encoder can extract a low-dimensional factor coordinate, a nonlinear network can estimate the factor score, and skip or linear connections can represent the complement score. The implementation is available at https://github.com/xymmmm00/diffusion_factor_model. The resulting statistical analysis depends primarily on the intrinsic factor dimension k , rather than only on the ambient number of assets d , which is the main advantage of the factor structure in high-dimensional and small-sample return generation.

3 Possible Directions

3.1 Heavy-Tailed Residuals, Flow Matching, and Guarantees

This direction keeps the same organizing principle as DFM: systematic variation should be represented through a low-dimensional factor space, while residual directions should remain structured enough to analyze. The main change is to relax the Gaussian residual or the Gaussian diffusion path so that the generated return distribution can better represent tail events.

3.1.1 Student- t Residuals and Score Decomposition

DFM relies on a Gaussian idiosyncratic residual. This assumption makes the score decomposition in (2) clean: after the OU perturbation, the residual component remains Gaussian with diagonal covariance $\mathbf{\Lambda}_t$, and the complement score becomes a linear function of $\mathbf{r}$. The limitation is that Gaussian residuals are light-tailed, meaning that the probability of very large idiosyncratic shocks decays quickly. Financial returns often contain extreme asset-specific moves, so a light-tailed residual model may underestimate tail risk even when the common factor structure is correct. This is also a broader concern for standard Gaussian diffusion models: Brownian perturbations and Gaussian denoising losses are natural and mathematically convenient, but they are not specifically designed to represent heavy-tailed targets or rare extreme events [Pandey et al., 2024, Yu and Yu, 2026, Yoon et al., 2023].

For example, we can replace the Gaussian residual with a multivariate Student- t residual while keeping the factor structure

$$\mathbf{R} = \beta \mathbf{F} + \epsilon, \quad \epsilon \sim t_\nu(\mathbf{0}, \mathbf{D}),$$

where $t_\nu(\mathbf{0}, \mathbf{D})$ denotes a d -dimensional Student- t distribution with degrees of freedom $\nu > 0$, location $\mathbf{0}$, and scale matrix $\mathbf{D}$. The parameter ν controls tail thickness: smaller ν gives heavier tails. The scale matrix is not the same as the covariance matrix; when $\nu > 2$, the residual covariance is $\nu(\nu - 2)^{-1}\mathbf{D}$.

The key reason this example is tractable is that a Student- t residual can be written as a conditional Gaussian mixture. Let π denote the prior law of a positive latent scale τ , and suppose

$$\epsilon \mid \tau \sim N(\mathbf{0}, \tau \mathbf{D}), \quad \tau \sim \text{Inv-Gamma}(\nu/2, \nu/2). \quad (3)$$

Here $\text{Inv-Gamma}(a, b)$ is parameterized by shape a and scale b . After integrating out τ , the marginal distribution of ϵ in (3) is $t_\nu(\mathbf{0}, \mathbf{D})$, giving the Student- t residual model above. Conditional on τ , however, the model is exactly a Gaussian-residual factor model, so the decomposition in (2) can be repeated with

$$\mathbf{\Lambda}_t(\tau) := h_t \mathbf{I}_d + \alpha_t^2 \tau \mathbf{D}, \quad \mathbf{\Gamma}_t(\tau) := (\beta^\top \mathbf{\Lambda}_t(\tau)^{-1} \beta)^{-1},$$

and

$$\mathbf{T}_t(\tau) = \mathbf{\Lambda}_t(\tau)^{-1/2} \beta \mathbf{\Gamma}_t(\tau) \beta^\top \mathbf{\Lambda}_t(\tau)^{-1/2}.$$

For this conditional Gaussian model, the score $\mathbf{s}_\tau^*(\mathbf{r}, t) := \nabla \log p_t(\mathbf{r} \mid \tau)$ has the form

$$\begin{aligned} \mathbf{s}_\tau^*(\mathbf{r}, t) = & \underbrace{\mathbf{T}_t(\tau) \mathbf{\Lambda}_t(\tau)^{1/2} \beta \nabla \log p_{\text{fac}, t}^{(\tau)} \left(\beta^\top \mathbf{\Lambda}_t(\tau)^{1/2} \mathbf{T}_t(\tau) \mathbf{\Lambda}_t(\tau)^{-1/2} \mathbf{r} \right)}_{\text{Conditional subspace score}} \\ & - \underbrace{\mathbf{\Lambda}_t(\tau)^{-1/2} (\mathbf{I}_d - \mathbf{T}_t(\tau)) \mathbf{\Lambda}_t(\tau)^{-1/2} \mathbf{r}}_{\text{Conditional complement score}}, \end{aligned} \quad (4)$$

where the conditional factor density is

$$p_{\text{fac}, t}^{(\tau)}(\mathbf{z}) := \int_{\mathbb{R}^k} \phi(\mathbf{z}; \alpha_t \mathbf{f}, \mathbf{\Gamma}_t(\tau)) p_{\text{fac}}(\mathbf{f}) d\mathbf{f}.$$

This density is the analogue of $p_{\text{fac}, t}$, with the conditional covariance $\mathbf{\Gamma}_t(\tau)$ replacing $\mathbf{\Gamma}_t$. The marginal noisy density is the mixture

$$p_t(\mathbf{r}) = \int p_t(\mathbf{r} \mid \tau) \pi(d\tau).$$

Therefore the marginal score can be written as

$$\mathbf{s}^*(\mathbf{r}, t) = \nabla \log p_t(\mathbf{r}) = \mathbb{E}[\mathbf{s}_\tau^*(\mathbf{r}, t) \mid \mathbf{R}_t = \mathbf{r}], \quad (5)$$

where the conditional expectation is taken with respect to the posterior law

$$\pi(d\tau \mid \mathbf{R}_t = \mathbf{r}) = \frac{p_t(\mathbf{r} \mid \tau) \pi(d\tau)}{p_t(\mathbf{r})}.$$

This identity follows from the derivative calculation

$$\nabla p_t(\mathbf{r}) = \int p_t(\mathbf{r} \mid \tau) \mathbf{s}_\tau^*(\mathbf{r}, t) \pi(d\tau).$$

The mixture density, posterior law, and derivative identity together give the posterior-average score formula in (5). Equations (4) and (5) suggest that a useful score decomposition may survive in the Student- t case, but in a weaker form: the marginal score is a posterior average of conditional diffusion factor decompositions. The complement score is no longer globally linear in $\mathbf{r}$, because the posterior distribution of τ and the conditional matrices all depend on the latent scale. Still, the formula preserves the factor-model geometry and gives a concrete starting point for architecture design and analysis.

Another possible example is an asset-specific scale mixture,

$$\boldsymbol{\varepsilon} \mid \boldsymbol{\tau} \sim N(\mathbf{0}, \text{diag}\{\tau_1 \sigma_1^2, \dots, \tau_d \sigma_d^2\}),$$

where $\boldsymbol{\tau} = (\tau_1, \dots, \tau_d)^\top \in \mathbb{R}^d$ is a vector of positive random scales. This model allows different assets to have different tail thicknesses. Conditional on $\boldsymbol{\tau}$, the score again has the same Gaussian form as (4), now with

$$\boldsymbol{\Lambda}_t(\boldsymbol{\tau}) := \text{diag}\{h_t + \alpha_t^2 \tau_1 \sigma_1^2, \dots, h_t + \alpha_t^2 \tau_d \sigma_d^2\}.$$

The marginal score is again a posterior average over $\boldsymbol{\tau}$, as in (5). This version is more flexible, but also harder to analyze because the projection matrix $\mathbf{T}_t(\boldsymbol{\tau})$ changes with the latent scale vector. A more aggressive possibility is an α -stable residual, whose tails can be much heavier and whose increments are related to Lévy-process diffusion models [Yoon et al., 2023]. However, when $\alpha < 2$, second moments may not exist, so covariance-based quantities and Gaussian-style denoising arguments need to be replaced by different tools.

3.1.2 Flow Matching as an Alternative Generator

Heavy-tailed residuals can also be approached through flow matching rather than score-based diffusion. Flow matching trains a time-dependent vector field that transports a simple initial distribution to the target data distribution; it can be viewed as a continuous normalizing flow trained by regression on prescribed probability paths [Lipman et al., 2024]. For a factor model, one could train a vector field on returns $\mathbf{R}$, or design an architecture that first maps $\mathbf{R}$ toward a low-dimensional factor coordinate and then transports both factor and residual components. Since flow matching does not require the same Brownian forward process or Gaussian denoising score as standard diffusion, it may be more flexible for heavy-tailed residuals.

This connects naturally with heavy-tailed priors in flow matching. Guan et al. [2025] study mirror flow matching with heavy-tailed priors for generative modeling on convex domains. In our setting, a related idea would be to choose a heavy-tailed base distribution for residual directions, or to use a flow path that preserves heavy-tailed behavior more directly than Gaussian noising. This would not automatically give the score decomposition in (2), because flow matching learns a velocity field rather than a score. Still, the same factor-model geometry can guide the architecture: one part of the vector field can act on a k -dimensional factor coordinate, while another part handles idiosyncratic residual directions.

3.1.3 Theoretical Goals

The analysis in DFM separates score approximation and estimation first, and distribution estimation second. We should follow the same structure for the heavy-tailed extension. The first goal is a fixed-time score estimation guarantee. For each $t \in [t_0, T]$, or uniformly over this interval, we would like to control

$$\mathcal{E}_{\text{score}}(t) := \mathbb{E}_{\mathbf{R}_t \sim P_t} \left[\|\hat{\mathbf{s}}(\mathbf{R}_t, t) - \mathbf{s}^*(\mathbf{R}_t, t)\|_2^2 \right], \quad (6)$$

where $\hat{\mathbf{s}}$ is the learned score network and $\mathbf{s}^*$ is the true score. This is the analogue of the $L^2(P_t)$ score risk used in the approximation analysis of DFM. For a heavy-tailed residual model, the main question is whether the bound for (6) can still depend primarily on the factor dimension k , plus a manageable measure of residual-tail complexity, rather than suffering a fully d -dimensional nonparametric rate. In the Student- t scale-mixture case, this would require controlling the conditional Gaussian score in (4) and the posterior averaging step in (5).

The second goal is a distribution estimation guarantee. This is where the time integration should enter. Let $\hat{P}_{t_0}$ be the law generated by the learned reverse process stopped at early time t_0 , and let P_{data} be the true return distribution. Following the analysis in DFM, one can decompose the total variation error into early-stopping error, score-estimation error, and terminal-mixing error:

$$\text{TV}(P_{\text{data}}, \hat{P}_{t_0}) \leq \text{TV}(P_{\text{data}}, P_{t_0}) + \text{TV}(P_{t_0}, \tilde{P}_{t_0}) + \text{TV}(\tilde{P}_{t_0}, \hat{P}_{t_0}).$$

Here P_{t_0} is the true forward-noised distribution at time t_0 , $\tilde{P}_{t_0}$ is the law of the reverse process driven by the learned score but initialized from the true terminal law P_T , and $\hat{P}_{t_0}$ is the practical generated law initialized from the standard Gaussian terminal distribution. The middle term is the place where a coupling argument converts the score error along the reverse trajectory into distributional error, through an integrated quantity such as

$$\int_{t_0}^T \mathbb{E}_{\mathbf{R}_t \sim P_t} \left[\|\hat{\mathbf{s}}(\mathbf{R}_t, t) - \mathbf{s}^*(\mathbf{R}_t, t)\|_2^2 \right] dt.$$

For scale-mixture residuals such as (3), the proof would need to combine the conditional Gaussian decomposition with concentration or coupling arguments over the latent scale variables. For a flow-matching approach, the analogous goal is to control the error between the learned vector field and the target vector field, and then translate that error into a bound on the generated distribution. In both cases, the ideal result would preserve the central message of DFM: high-dimensional return generation should be governed mainly by the intrinsic factor dimension and the residual structure, not merely by the ambient number of assets.

3.2 Learning Factors Through Score Networks

The second direction does not require changing the baseline model in DFM. We keep the factor model (1), the Gaussian idiosyncratic residual, and the OU diffusion process. The new question is not how to generate high-dimensional returns, but how much factor information is encoded in the learned score network after it has been trained on returns. The observed data are still $\{\mathbf{r}_i\}_{i=1}^n$, not paired observations of factors or residuals, so the target must respect the usual identifiability limitation of factor models: individual factor coordinates are not unique, while factor spaces and systematic return components are meaningful up to rotation and scaling.

We start with a simplified version of factor model, but introduce a period index to the asset returns:

$$\mathbf{R}_i = \beta_i \mathbf{F}_i,$$

where i is the economic period, $\mathbf{R}_i \in \mathbb{R}^d$ is the noiseless return vector of d assets, $\mathbf{F}_i \in \mathbb{R}^k$ is the vector of k latent factors, and $\beta_i \in \mathbb{R}^{d \times k}$ is the period-specific loading matrix. Under the context of practically used factor models, factor exposure consists of descriptors and gives the exposure matrix β_i the formulation

$$\beta_i = \mathbf{U}_i \mathbf{T},$$

where $\mathbf{U}_i \in \mathbb{R}^{d \times m}$ is a pre-computed matrix of m asset descriptors or fundamentals, such as valuations, growth metrics, or risk attributes. The weight matrix $\mathbf{T} \in \mathbb{R}^{m \times k}$ is fixed, typically derived from domain expertise, and *invariant over time*.

Denote the thin QR-decomposition of $\mathbf{U}_i$ as $\mathbf{U}_i = \mathbf{Q}_i \mathbf{V}_i$, where $\mathbf{Q}_i \in \mathbb{R}^{d \times m}$ has orthonormal columns and $\mathbf{V}_i \in \mathbb{R}^{m \times m}$ is upper triangular. With some algebra, we compute the score function of asset returns at period i and diffusion time t as

$$\nabla \log p_{i,t}(\mathbf{r}) = -\frac{1}{\sigma_t^2} \mathbf{r} - \frac{\alpha_t}{\sigma_t^2} \mathbf{U}_i \mathbf{T} \frac{\int \mathbf{f}_i \exp\left(-\frac{\|\mathbf{Q}_i^\top \mathbf{r} - \alpha_t \mathbf{V}_i \mathbf{T} \mathbf{f}_i\|_2^2}{2\sigma_t^2}\right) p_f(\mathbf{f}_i) d\mathbf{f}_i}{\int \exp\left(-\frac{\|\mathbf{Q}_i^\top \mathbf{r} - \alpha_t \mathbf{V}_i \mathbf{T} \mathbf{f}_i\|_2^2}{2\sigma_t^2}\right) p_f(\mathbf{f}_i) d\mathbf{f}_i},$$

where σ_t^2 is the scalar Gaussian noising variance at diffusion time t , $\mathbf{r} \in \mathbb{R}^d$, and $\mathbf{f}_i \in \mathbb{R}^k$. We write p_f for the underlying factor density in this simplified setting. Dimensionally, $\mathbf{Q}_i^\top \mathbf{r} \in \mathbb{R}^m$, $\mathbf{V}_i \mathbf{T} \mathbf{f}_i \in \mathbb{R}^m$, and the integral ratio lies in $\mathbb{R}^k$. Corresponding to the score function, we choose the following neural network to estimate the score:

$$\mathbf{s}_\theta(\mathbf{r}, t, i) = -\frac{1}{\sigma_t^2} \mathbf{r} - \frac{\alpha_t}{\sigma_t^2} \mathbf{U}_i \mathbf{A} \cdot \text{Network}(\mathbf{Q}_i^\top \mathbf{r}, t, i).$$

Here $\mathbf{A} \in \mathbb{R}^{m \times k}$ is a trainable weight matrix estimating the time-invariant matrix $\mathbf{T}$, and $\text{Network}(\mathbf{Q}_i^\top \mathbf{r}, t, i) \in \mathbb{R}^k$ is a deep neural network output. Therefore $\mathbf{U}_i \mathbf{A} \in \mathbb{R}^{d \times k}$, and the whole estimator $\mathbf{s}_\theta(\mathbf{r}, t, i)$ lies in $\mathbb{R}^d$, matching the dimension of the score.

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